

Dealing with Unpredictability

How can we describe unpredictability, uncertainty and randomness?

Researchers aim to understand the way the world works on the basis of incomplete information

- Sometimes we don't know what we need to measure
- Sometimes we know what we need to measure but we can't measure it
- Sometimes we know what we need to measure, we can measure it, but we don't have the resources to measure everything

Incomplete information introduces

- Uncertainty and unpredictability, aka randomness

What ingredients do we need to describe randomness?

1. A sample space Ω

– The set of possible outcomes of an experiment

- Coin toss: $\Omega = \{H, T\}$
- Two coin tosses: $\Omega = \{HH, HT, TH, TT\}$
- Assessment grades: $\Omega = \{1, 2, 3, 4, 5, 6, 7\}$
- Lengths of thumbs: $\Omega = \mathbf{R}_+ = [0, \infty)$
- Lengths of thumbs and index fingers:
 $\Omega = \mathbf{R}_+^2 = (0, \infty) \times (0, \infty)$

What ingredients do we need to describe randomness?

2. Events

- A subset of the sample space
 - Coin toss: $\Omega = \{H, T\}$
 - The outcome is a head
 - The outcome is a tail

What ingredients do we need to describe randomness?

2. Events

– A subset of the sample space

- Two coin tosses: $\Omega = \{HH, HT, TH, TT\}$
- The first toss is heads
 $A = \{HH, HT\}$
- No tails
 $A = \{HH\}$
- The tosses are different
 $A = \{HT, TH\}$

What ingredients do we need to describe randomness?

2. Events

– A subset of the sample space

- Assessment grades: $\Omega = \{1, 2, 3, 4, 5, 6, 7\}$

- You get a 7

$A = \{7\}$

- You pass

$A = \{4, 5, 6, 7\}$

What ingredients do we need to describe randomness?

2. Events

– A subset of the sample space

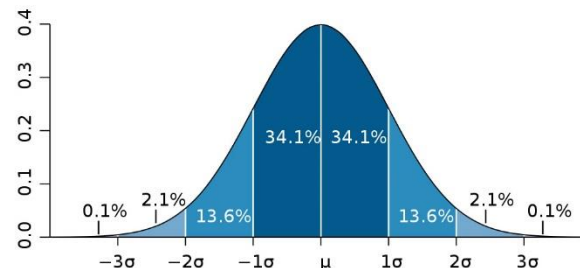
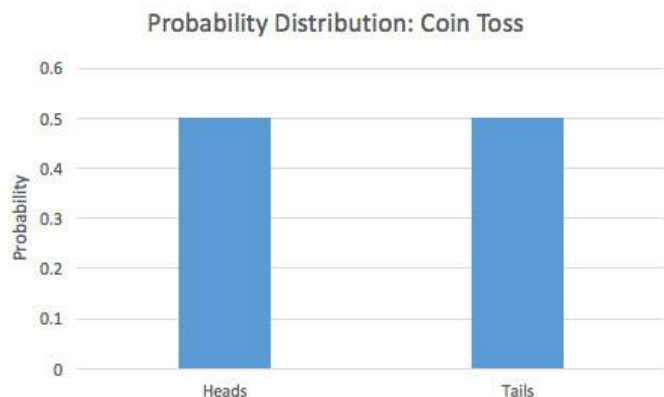
- Lengths of thumbs: $\Omega = \mathbf{R}_+ = (0, \infty)$
- Your thumb is $> 50\text{mm}$ and $\leq 60\text{mm}$
 $A = \text{thumb in } (50, 60]$
- Lengths of thumbs and index fingers:
 $\Omega = \mathbf{R}_+^2 = (0, \infty) \times (0, \infty)$
- Your thumb is $> 50\text{mm}$ and $\leq 60\text{mm}$
AND
Your index is $> 80\text{mm}$ and $\leq 90\text{mm}$
 $A = \text{thumb in } (50, 60] \text{ and index in } (80, 90]$

What ingredients do we need to describe randomness?

3. A probability density function P

– Assigns a real number to every event so that

- $P(A) \geq 0$ for every event
- $P(\Omega) = 1$
- Can be discrete or continuous
- If events $A_1, A_2, \dots$ are disjoint then
$$P(A_1 \text{ or } A_2 \text{ or } \dots) = P(A_1) + P(A_2) + \dots$$



What ingredients do we need to describe randomness?

- A probability distribution has
 1. An outcome space
 2. Events that are subsets of that space
 3. A probability distribution function that assigns probabilities to those events

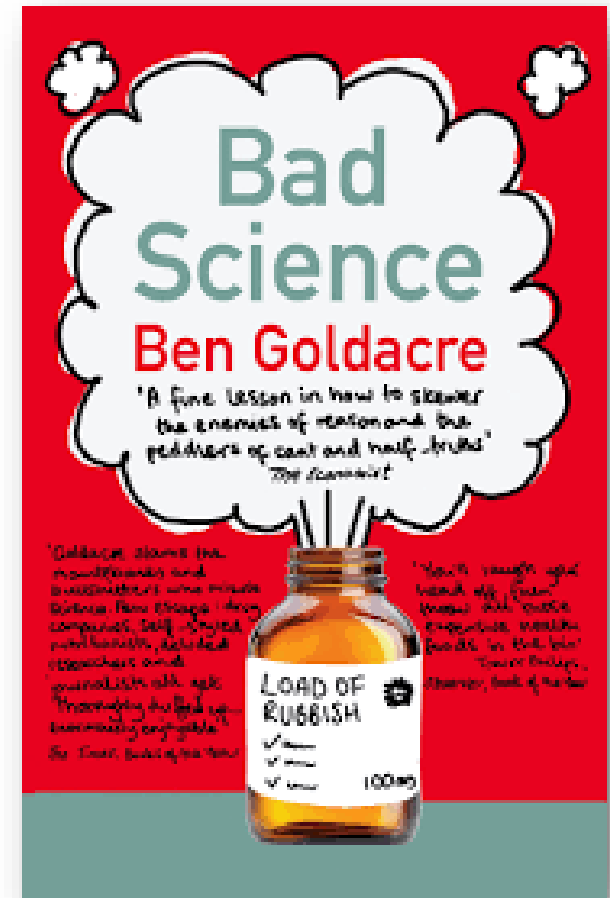
What are independent events?

(And how can we be misled
by assumptions of independence?)

- Two events are independent if
 - the outcome of one event has no effect on the outcome of the other
- If events A and B are independent
 - the probability of A and B happening together is $P(AB)=P(A)P(B)$

Independence assumption and SIDS

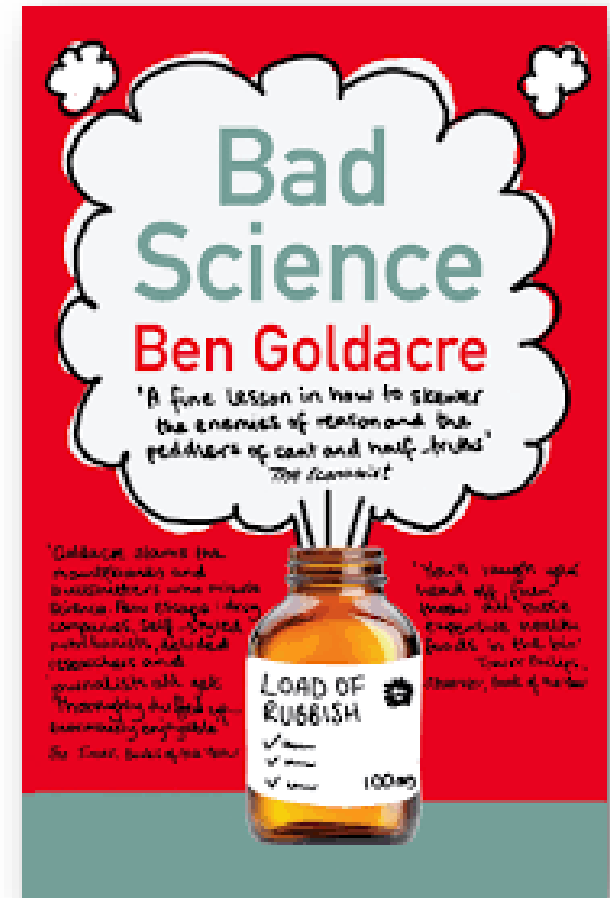
- 11-week old Christopher Clark, son of Sally and Steve died in his sleep
- A year later, the Clark's second child Harry died aged 8 weeks
- Sally and Steve were arrested
- Sally was sentenced to life in prison for the murder of Christopher and Harry



See Goldacre (2009) Bad Science, Chapter 14 Bad Statistics

Independence assumption and SIDS

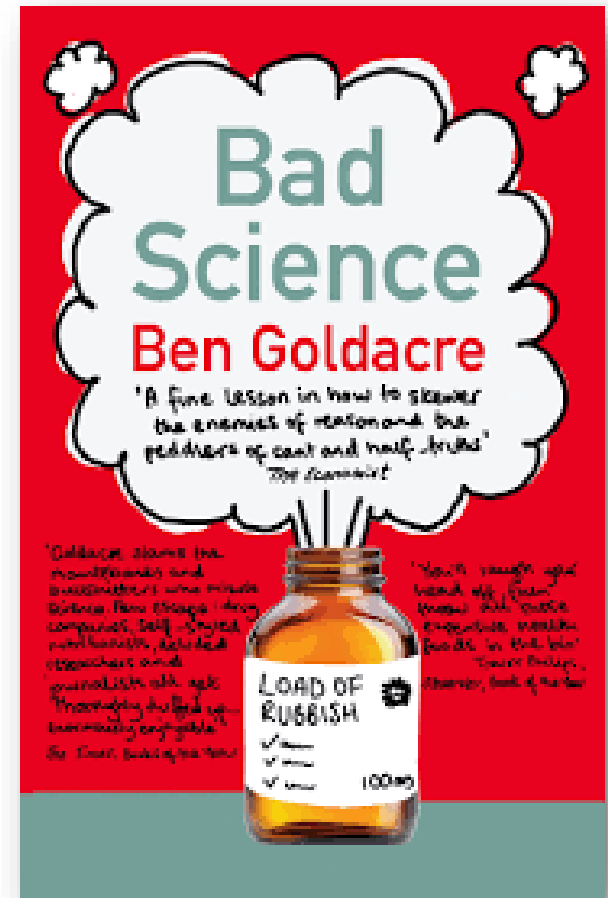
- Estimated chance of randomly chosen baby dying of SIDS: 1/8500
- Assuming that SIDS in first and second born babies is statistically independent, chances of two SIDS: $(1/8500)^2$
- 1 in 73 million



See Goldacre (2009) Bad Science, Chapter 14 Bad Statistics

Independence assumption and SIDS

- Later shown that second cases of SIDS are *not independent* of first cases of SIDS
- Couples whose firstborn dies of SIDS have around a 1/100 chance of their second born dying the same way
- incidence of double SIDS ~ 1 in 130,000
- See [Beyond reasonable doubt](#)



See Goldacre (2009) *Bad Science*, Chapter 14 *Bad Statistics*

Independence assumption and game shows

- Suppose you're on a game show, and you're given the choice of three doors.
 - Behind one door is a car, behind the others, goats.
 - You pick a door, say #1, and the host, who knows what's behind the doors, opens another door, say #3, which has a goat.
 - He says to you, "Do you want to pick door #2?"



Independence assumption and game shows

- Suppose you're on a game show, and you're given the choice of three doors.
 - Behind one door is a car, behind the others, goats.
 - You pick a door, say #1, and the host, who knows what's behind the doors, opens another door, say #3, which has a goat.
 - He says to you, "Do you want to pick door #2?"
- Have you ever seen this problem before?
 - Yes, I remember the answer
 - Yes but I forgot the answer
 - No
- Is it to your advantage to switch your choice of doors?
 - Yes; it is to your advantage
 - No; it is to your disadvantage
 - It makes no difference

Independence assumption and game shows

- Yes; you should switch
 - The first door (you pick) has a $1/3$ chance of winning
 - The second door (you could switch to) has a $2/3$ chance
- [Marilyn Vos Savant's game show problem](#)
 - Consider how many Professors and PhD graduates were certain, (but wrong) about the answer

Independence

- Many engineering examples where independence is assumed
- Eg: Failure modes / reliability of system.
- Be critical!

How can we be fooled by conditional probabilities?

- If $P(B) > 0$
then the conditional probability of A given B is

$$P(A | B) = \frac{P(AB)}{P(B)}$$

Conditional probability
of A given B

Joint probability of A and B

Unconditional probability of B

- Your engineering firm has developed a test T for damage D in a component
 - The test has two outcomes
 - T+ indicating that the test diagnoses damage
 - T- indicating that the test does not diagnose damage
 - In reality, the component can be
 - D+ damaged
 - D- not damaged

- The probability
 - That a component is damaged is 1%
 - That the test correctly diagnoses damage is 80%
 - That the test misdiagnoses damage is 9.6%
- What is the probability that a component diagnosed as damaged is actually damaged?
- 0%-30% 30%-60% 60%-100%

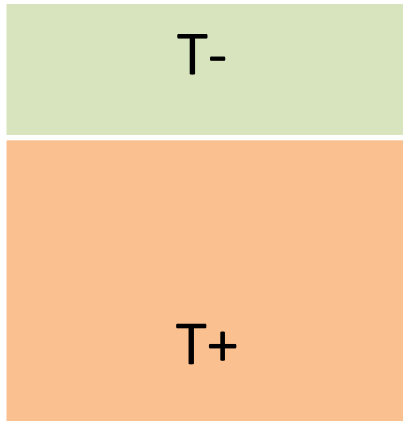
Probability that
there really is damage,
and
that the test is positive

Probability that
there is damage,
given that
the test is positive

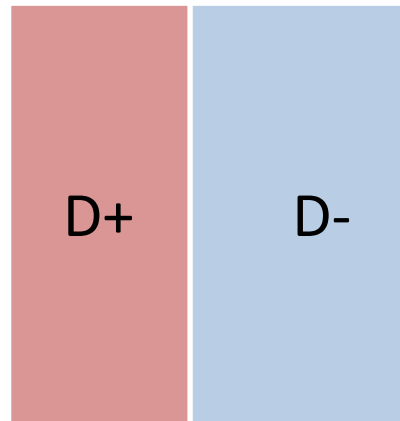
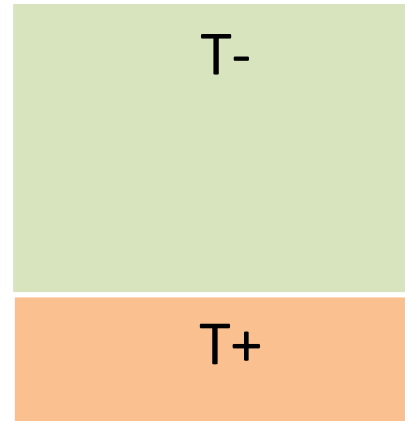
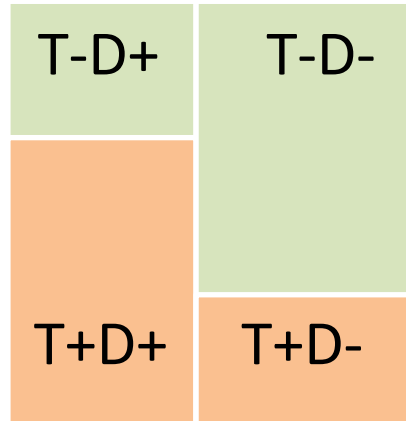
$$P(D+ | T+) = \frac{P(D+T+)}{P(T+)}$$

Probability that
the test is positive
(whether there is damage or not)

For damaged
components D+



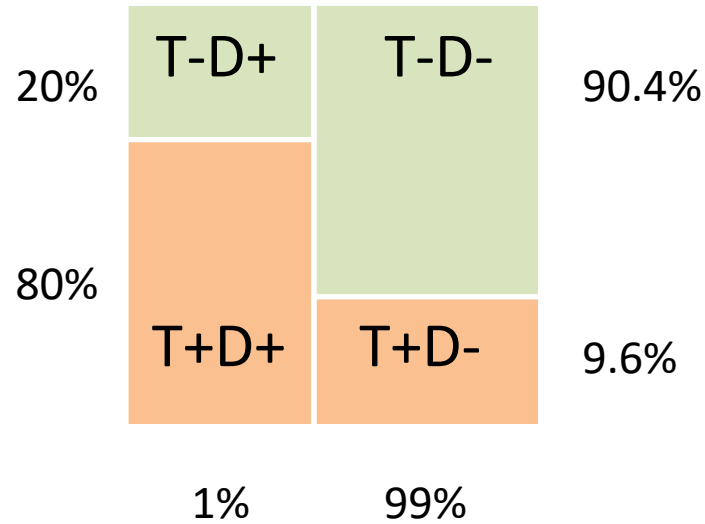
For undamaged
components D-



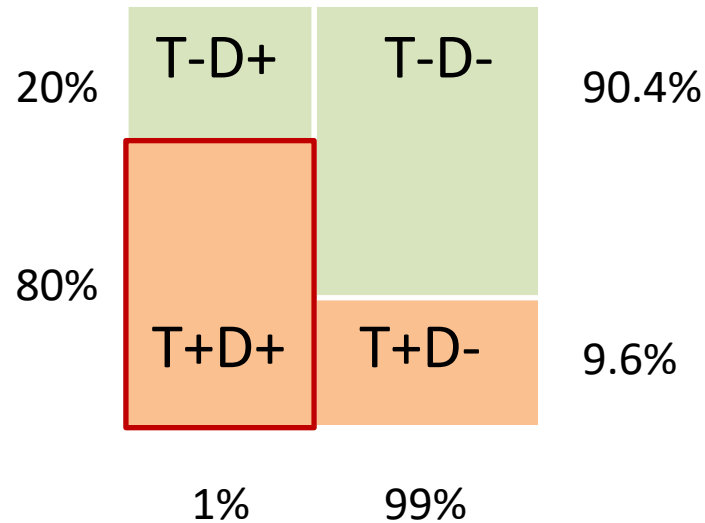
For all components

False negative	T-D+	T-D-	True negative
True positive	T+D+	T+D-	False positive

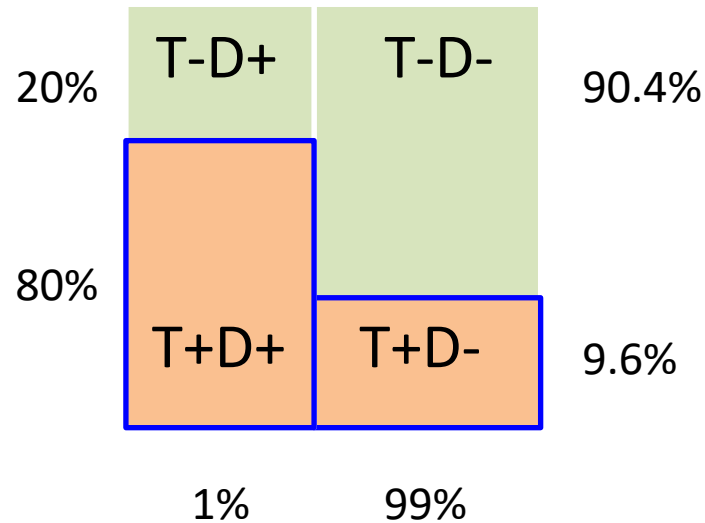
$$P(D+ | T+) = \frac{P(D+T+)}{P(T+)}$$



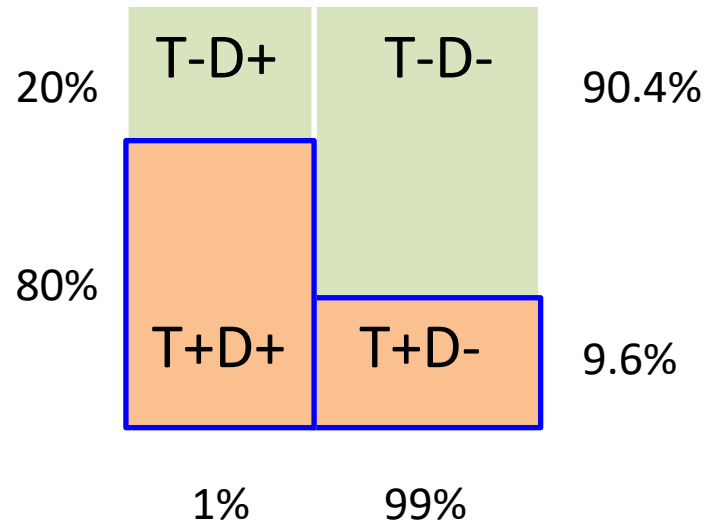
$$P(D+ | T+) = \frac{P(D+T+)}{P(T+)}$$



$$P(D+ | T+) = \frac{0.8 \times 0.01}{P(T+)}$$



$$P(D+ | T+) = \frac{0.8 \times 0.01}{0.8 \times 0.01 + 0.096 \times 0.99}$$



$$P(D+ | T+) = 0.078$$

- The probability
 - That a component is damaged is 1%
 - That the test correctly diagnoses damage is 80%
 - That the test misdiagnoses damage is 9.6%
- What is the probability that a component diagnosed as damaged is actually damaged?
- 7.8%

What's going on?

- This is an example of “base rate neglect”
 - The event we are trying to detect is rare
 - In this setting, the test returns far more false positives than true positives
- Here's an [area-proportional Euler diagram visualisation from Luana Micallef](#)
 - This time using a medical diagnosis example